

Jean-Sauveur Ay

Contact Information

Email: jean-sauveur.ay-at-inrae-dot-fr
Address: [UMR Cesaer, 26 blvd Dr Petitjean, 21000 Dijon \(France\)](#)
Web pages: [GitHub](#), [Lab](#), [RePEc](#), [ResearchGate](#), [Scholar](#)

Research interest

My research combines **economic theory** and **empirical methods** to generate evidence that supports **private decisions** and **public policies**. I specialize in **wine economics** and enjoy collaborating across disciplines to address **agricultural**, **environmental**, and **urban** outcomes.

Professional experience

2014 to now	Research scientist at INRAE, UMR Cesaer , Dijon
Spring 2026	Visiting scholar at Bren School , UC Santa Barbara
2013 to 2014	Research engineer at INRAE, UMR PSAE , Paris
2011 to 2013	Postdoctoral fellow at CNRS-MNHN, UMR Cescos , Paris
Summer 2006	Research internship at INRAE, UR Ecodev , Avignon

Education

July 2026	Accreditation to Supervise Research (HDR) at UBE
2008 to 2011	PhD in Economics at Université Bourgogne Europe (UBE)
2007 to 2008	M.Sc. Analysis and Policy in Economics at Paris School of Economics
2005 to 2008	Magistère Ingénieur Économiste at Aix-Marseille School of Economics

Current papers

- **The signaling values of nested wine names** (with Julie Le Gallo) [[wp](#)]
- **Private and social efficiency of insecticide use against an insect-borne infectious disease** (with Estelle Gozlan and Emmanuel Paroissien) [[file](#)]
- **Return on investment for promoting homeownership: The case of the French interest-free loan policy** (with Rémi Lei and Julie Le Gallo) [[file](#)]
- **Does urban sprawl cause residential vacancy across French municipalities?** (with Mahamoudou Zore and Catherine Baumont) [[file](#)]

Publications

Refereed papers (in English)

- **The dynamic impact of land-use changes on soil organic carbon** (with Liang Diao and Valentin Bellassen), *Environmental Research Letters*, 2026, 21, 184003 [[bib](#) ; [link](#) ; [file](#)]
- **Housing wealth beyond prices: Individual portfolios, distant assets, and transfers in France 2011–2022** (with Rémi Lei, Julie Le Gallo, and Olivier Dupré), *Review of Income and Wealth*, 2026 [[bib](#) ; [link](#)]
- **Strategic free-riding in pest control: Theory and evidence in organic-conventional mixed landscapes** (with François Bareille and Vincent Martinet), *Journal of Environmental Economics and Management*, 2026, 138 [[bib](#) ; [link](#)]
- **The social stratification of homeowners' housing wealth: Bringing a dynamic approach through the price gap index.** (with Thibault Le Corre) *Environment and Planning B: Urban Analytics and City Science*, 2026, 53(1): 523–542 [[bib](#) ; [file](#) ; [link](#) ; [online](#)]
- **External costs of high-voltage overhead transmission lines: Evidence from a causal approach in rural France** (with Tina Rambonilaza and Jean Dubé), *Environmental Impact Assessment Review*, 2024, 109(107639) [[bib](#) ; [file](#) ; [link](#)]
- **Building integrated plant health surveillance: a proactive research agenda for anticipating and mitigating disease and pest emergence** (with S. Soubeyrand, A. Estoup, G. Thébaud, and C. E. Morris and others). *CABI Agriculture and Bioscience*, 2024, 5(1), 72. [[bib](#) ; [link](#)]
- **Climate-dependent scenarios of land use for biodiversity and ecosystem services in the New Aquitaine region** (with N. A. Andriamanantena, C. Gaufreteau and L. Doyen), *Regional Environmental Change*, 2022, 22(3): 1–15 [[bib](#) ; [file](#) ; [sup](#) ; [link](#)].
- **Field and landscape risk factors impacting Flavescence dorée infection: Insights from spatial Bayesian modelling in the Bordeaux vineyards** (with H. K. Adrakey, S. Malembic-Maher, A. Rusch, L. Riley, L. Ramalanjaona and F. Fabre), *Phytopathology*, 2022, 112(8): 1686–1697 [[bib](#) ; [file](#) ; [link](#)]
- **The informational content of geographical indications**, *American Journal of Agricultural Economics*, 2021, 103(2): 523–542 [[bib](#) ; [file](#) ; [sup](#) ; [online](#) ; [link](#)]
- **Willingness-to-pay for reshuffling geographical indications** (with Monia Saïdi, Stéphan Marette and Christophe Martin), *Journal of Wine Economics*, 2020, 15(1): 95–111 [[bib](#) ; [file](#) ; [sup](#) ; [online](#) ; [link](#)]
- **Disease dispersion as a spatial interaction: The case of Flavescence Dorée** (with Estelle Gozlan), *Natural Resource Modeling*, 2020, 33:e12265 [[bib](#) ; [file](#) ; [pres](#) ; [link](#)]
- **Bioeconomic impacts of agroforestry policies in France** (with Lauriane Mouysset, Claire Rais Assa, Frédéric Jiguet, Romain Lorrilière, Luc Doyen), *Land Use Policy*, 2019, 85: 239–248. [[bib](#) ; [link](#) ; [wp](#)]
- **Does issuing building permits reduce the cost of land? An estimation based on the demand for building land in France** (with Jean Cavaillès, Mohamed Hilal and Julie Le Gallo), *Economics and Statistics*, 2018, 500-501-502, 45–67. [[bib](#) ; [file](#) ; [sup](#) ; [link](#)]

- **Nonlinear impact estimation in spatial autoregressive models** (with Kassoum Ayouba and Julie Le Gallo), *Economics Letters*, 2018, 163: 59–64. [[bib](#) ; [file](#) ; [sup](#) ; [link](#)]
- **Distance decay in the willingness to pay for wine: Disentangling local and organic attributes** (with Raja Chakir, and Stéphane Marette), *Environmental and Resource Economics*, 2017, 68(4): 997–1019. [[bib](#) ; [file](#) ; [pres](#) ; [sup](#) ; [link](#)]
- **Aggregated versus individual land-use models: Modeling spatial autocorrelation to increase predictive accuracy** (with Raja Chakir and Julie Le Gallo), *Environmental Modeling and Assessment*, 2017, 22(2): 129–145. [[bib](#) ; [file](#) ; [sup](#) ; [link](#) ; [online](#) ; [wp](#)]
- **The economics of land use reveals a selection bias in tree species distribution models** (with Joannès Guillemot, Nicolas Martin-St Paul, Luc Doyen, and Paul Leadley), *Global Ecology and Biogeography*, 2017, 26(1):65–77. [[bib](#) ; [file](#) ; [pres](#) ; [sup](#) ; [link](#)]
- **The informational content of land price and its relevance for environmental issues** (with Laure Latruffe), *Review of Environmental and Resource Economics*, 2016, 10(3-4):183–226. [[bib](#) ; [file](#) ; [link](#) ; [wp](#)]
- **Integrated models, scenarios and dynamics of climate, land use and common birds** (with Raja Chakir, Luc Doyen, Frédéric Jiguet and Paul Leadley), *Climatic Change*, 2014, 126(1-2):13–30. [[bib](#) ; [file](#) ; [sup](#) ; [link](#) ; [online](#)]
- **Efficiency and equity of land conservation: The effect of policy scale** (with Claude Napoléone), *Journal of Environmental Management*, 2013, 129:190–198. [[bib](#) ; [link](#)]
- **A spatial hedonic approach on land use change anticipations** (with Ghislain Geniaux and Claude Napoléone), *Journal of Regional Science*, 2011, 51(5):967–986. [[bib](#) ; [link](#)]

Refereed papers (in French)

- **Mises en oeuvre de démarches agroécologiques dans des régions viticoles françaises** (avec Patrice This, Emna Troudi, Marie Thiollot-Scholtus et autres), *Innovations Agronomiques*, 2025. [[bib](#) ; [link](#)]
- **La capitalisation du prix des vins dans le prix des vignes en Côte-d'Or** (avec Abdoul Diallo et Hai-Vu Pham), *Revue Économique*, 2023, 74 (1). [[bib](#) ; [file](#)]
- **Les déterminants naturels et politiques des AOC viticoles de Côte-d'Or** (avec Mohamed Hilal), *CyberGeo*, 2021. [[bib](#) ; [file](#) ; [sup](#) ; [link](#)]
- **Vers une évaluation des coûts de la dégradation des sols : Éléments de cadrage, outil d'analyse et études de cas** (avec Noémie Pousse, Laurent Rigou et Laurent Thannberger), *Étude et Gestion des Sols*, 2020, 27:147–161. [[bib](#) ; [file](#) ; [link](#)]
- **Délivrer des permis de construire pour diminuer le coût du foncier ? Une estimation par la demande de terre constructible** (avec Jean Cavaillès, Mohamed Hilal et Julie Le Gallo), *Économie et Statistiques*, 2018, 500-501-502, 45–67. [[file](#) ; [sup](#) ; [link](#) ; [pres](#)]
- **Information sur l'hétérogénéité de la terre et délégation de la régulation foncière**, *Revue d'Économie Politique*, 2015, 125(3):453–474. [[bib](#) ; [link](#) ; [wp](#)]

- **La gestion publique des ressources foncières: Entre évaluation de la rareté et choix d'usage** (avec Claude Napoléone), *Revue des Régions Arides*, 2013, 30(1):177–190. [[bib](#) ; [file](#)]

Other publications

- **Préserver la qualité des sols: Vers un référentiel d'indicateurs.** (avec Isabelle Cousin, Maylis Desrousseaux, Sophie Leenhardt et autres). *Expertise collective INRAE*, 2025, [[link](#)]
- **Les consommateurs face à l'évolution des Appellations d'Origine Contrôlée** (avec Monia Saïdi, Stéphan Marette and Christophe Martin). *INRAE Sciences Sociales*, 2020, [[link](#)]
- **The drivers of spatial cropping patterns in Burgundy** (with Valentin Bellassen and Mohamed Hilal). *UMR CESAER Working Papers 2020-1*, 2020, [[link](#)]
- **Sols artificialisés et processus d'artificialisation des sols: Déterminants, impacts et leviers d'action** (avec Béatrice Béchet, Yves Le Bissonnais, Anne Ruas, Bertrand Schmitt et autres) *Expertise collective INRA-IFSTTAR*, 2017, [[link](#)]
- **Improving the estimates of agricultural land valuation: Report to the World Bank** (with Philippe Bontems, Raja Chakir and Laure Latruffe). *HAL-01462707*, 2015 [[link](#)]
- **La valeur des attributs naturels des terres agricoles de Côte-d'Or** (with Jean-Marc Brayer, Jean Cavailhès, Pierre Curmi, Mohamed Hilal and Marjorie Ubertos). *CESAER Working Paper*, 2012, [[link](#)]
- **Hétérogénéité de la terre et rareté économique.** *PhD thesis, Université de Bourgogne Europe*, 2011, [[link](#)]

Communications

- 2026 emLab seminar** (April, Santa Barbara, USA) ; **Acclimate workshop** (September, Regnéville-sur-Mer, France).
- 2025 Jeudis Expertise Vigne & Vin** (April, Dijon, France) ; **Colloque OMISEP** (June, Marseille, France) ; **Conseil scientifique FNSAFER** (October, Paris, France) ; **Conférence Chaire UNESCO** (November, Dijon, France) ; **Séminaire FAR** (November, Paris, France).
- 2024 Séminaire FAR** (June, Paris, France) ; **Séminaire INED** (June, Paris, France) ; **Final Meeting SERENA** (October, Orléans, France) ; **Journées de Recherches en Sciences Sociales** (December, Reims, France).
- 2023 AERE@AEE** (February, New York, USA) ; **Spatial Econometrics Workshop** (May, Dijon, France) ; **Journées PSAE** (June, Paris, France) **Journées d'Étude des Sols** (June, Dijon, France) ; **Jeudis Expertise Vigne & Vin** (September, Dijon, France).
- 2022 Séminaire FAR** (March, Paris, France) ; **Wine and Spirit Seminar of the Burgundy School of Business** (June, Dijon, France) ; **Association Française de Science Économique** (June, Dijon, France)

- 2021 Journées INRAE–SFER–CIRAD** (March, Angers, France) ; **Atelier Vin et Société at Institut d'études avancées** (April, Paris, France) ; **Séminaire SMART-LERECO** (October, Rennes, France) ; **Journées Recherches en Sciences Sociales** (December, Toulouse, France)
- 2020 Séminaire Vigne et Vins** (March, Dijon, France).
- 2019 World Conference on Natural Resource Modeling** (May, Montréal, Canada) ; **Transmission du Patrimoine Viticole en Bourgogne** (June, Beaune, France)
- 2018 Séminaire LARESS à l'ESA** (April, Angers, France) ; **Journées Ingé plus** (April, Dijon, France).
- 2017 Ministère de l'écologie** (March, Paris, France) ; **Séminaire du LEF** (March, Nancy, France) ; **Vineyard Data Quantification Society** (May, Colmar, France) ; **Spatial Econometrics Workshop** (May, Avignon, France) ; **Société Française d'Écologie** (October, Marseille, France) ; **Séminaire du LAMETA** (December, Montpellier, France) ; **Journée du département SAE2** (December, Paris, France).
- 2015 Salon International de l'Agriculture** (February, Paris, France) ; **Wine seminar at Kedge School of Business** (May, Bordeaux, France) ; **American Association of Wine Economists** (May, Mendoza, Argentina) ; **Natural Resource Modelling workshop** (June, Bordeaux, France) ; **TRUSTEE General Meeting** (October, Malmö, Sweden) ; **Dialogues Interdisciplinaires**, (December, Dijon, France).
- 2014 Global Land Project Science meeting** (March, Berlin, Germany) ; **Métaprogramme INRA ACCAF** (December, Orléans, France).
- 2013 VII Conference of Spatial Econometrics Association** (July, Washington, USA) ; **Internal Seminar CRESE** (October, Besançon, France) ; **Global Change Adaptation** (November, Clermont-Ferrand, France) ; **Journées INRA–SFER–CIRAD** (December, Angers, France).
- 2012 Planning, Law and Property Rights** (February, Belfast, United Kingdom) ; **Workshop Spatial Econometrics and Statistics** (November, Avignon, France) ; **Journées INRA–SFER–CIRAD** (December, Toulouse, France).
- 2010 Internal Seminar Ecodéveloppement** (October, Avignon, France).
- 2009 EAAE Ph.D. Workshop** (September, Giessen, Germany) ; **Journées d'Étude des Sols** (May, Strasbourg, France).
- 2008 Journées INRA–SFER–CIRAD** (December, Lille, France).

Other activities

Research programs

- MORSUR (2026-2029), *Région Bourgogne-Franche-Comté*, WP leader;
- [RURALITIC](#) (2025–2028) Horizon Europe, Participant;
- [OPOSOM](#) (2024–2026) *Agence Nationale de la Recherche*, WP leader;
- [SolFermage](#) (2024–2026) *Institut Robert Badinter*, WP leader;

- [LAMARTINE](#) (2023–2026) *Agence Nationale de la Recherche*, Participant;
- [Risca 2](#) (2023–2026) *Plan National Dépérissement du Vignoble*, WP leader;
- [ACCLIMATE](#) (2022–2026) *Agence Nationale de la Recherche*, Participant;
- [SERENA](#) (2022–2025) European Joint Program, Participant;
- [LOCUS](#) (2022–2025) *Agence Nationale de la Recherche*, WP leader;
- [OMISEP](#) (2022–2024) *Agence de la Transition Écologique*, WP leader;
- [FAST](#) (2021–2026) *Agence Nationale de la Recherche*, Participant;
- [Risca](#) (2019–2022) *Plan National Dépérissement du Vignoble*, WP leader;
- [PubPrivLands](#) (2019–2021) *Région Bourgogne-Franche-Comté*, WP leader;
- [Trustee](#) (2013–2017) ERA-NET RURAGRI program, Participant;
- [Mobilis](#) (2012–2013) *Fondation pour la Recherche sur la Biodiversité*, Participant;
- [EcoSolHydro](#) (2011–2013) *Agence de la Transition Écologique*, Participant.

Teaching experience

Course	Place	h/yr	Level	Period
Econometrics	Sciences Po Dijon	24	2nd year	2018–2027 (9 yrs)
Economics	Sciences Po Dijon	48	1st year	2016–2021 (5 yrs)
Wine Economics	Institut Agro Dijon	3	3rd year	2016, 2026 (2 yrs)
Land Economics	Univ. of Franche-Comté	6	Master 1	2023 (1 yr)
Econometrics	AgroParisTech	24	Master 1	2012–2017 (5 yrs)
Econometrics	Univ. of Franche-Comté	18	Master 1	2012–2017 (5 yrs)
Microeconomics	Univ. Bourgogne Europe	14	License 2	2010–2011 (1 yr)

Referee reports

Acta Oeconomica (1), Agricultural Economics (1), American Journal of Agricultural Economics (2), Applied Economics (1), Ecological Economics (2), Économie et Statistique (2), Économie Rurale (3), Environmental Modeling and Assessment (2), Environment and Planning A (1), Environmental and Resource Economics (1), European Review of Agricultural Economics (3), Geoderma (1), International Journal of Geographical Information Science (1), International Journal of Strategic Property Management (1), Journal of Agricultural Economics (1), Journal of Environmental Management (3), Land Economics (1), OENO One (1), Plos One (1), Regional Studies (1), Review of Agricultural, Food and Environmental Studies (2), Revue Économique (1), Revue d'Économie Régionale et Urbaine (3), Spatial Economic Analysis (2), Spatial Statistics (1), Sustainability (1).

Coordination and expertise

- **Since 2026:** coordinator of the FAR network (agricultural and rural land) at INRAE,

- **Since 2024:** member of the council of the French association of agricultural economists,
- **Since 2021:** member of the working group on the wine and grapevine products at INRAE,
- **Since 2014:** consultant and expert for the French National Institute of Origin and Quality,
- **Since 2014:** head of the CESAER scientific seminar,
- **2024–2026:** INRAE collective scientific assessment on soil quality indicators,
- **2024:** construction of individual housing-wealth data at CEREMA,
- **2018–2024:** member of the national network of scientific and technical expertise on soils,
- **2022:** organization of the annual congress of the French Economic Association in Dijon,
- **2021:** hearing by the INRAE / CIRAD / IRD joint ethics committee on soils,
- **2016–2018:** member of the scientific committee of the Workshop on Spatial Econometrics,
- **2016, 2018, 2024:** initiation and organization of the CESAER lab research days,
- **2014–2016:** INRAE collective assessment on artificialized land and artificialization processes,
- **2012–2014:** consultant for Inter-Rhône, the wine trade organization of the Rhône Valley,
- **2009–2011:** initiation and organization of the CESAER internal working group,
- **Other:** Involved in [Distillerie Mazy](#) in Burgundy and [Domaine Raspail-Ay](#) in Rhône Valley.